

well as for the entire model, when running on a Nvidia H100 GPU with BF16 precision. DyT layers reduce computation time compared to RMSNorm layers.

LLaMA 7B	inference		training	
	layer	model	layer	model
RMSNorm	2.1s	14.1s	8.3s	42.6s
DyT	1.0s	13.0s	4.8s	39.1s
reduction	↓52.4%	↓7.8%	↓42.2%	↓8.2%

Table 14 Inference and training latency (BF16 precision) for LLaMA 7B with RMSNorm or DyT. DyT achieves a substantial reduction in both inference and training time. Results are measured without any extra performance optimizations.

We also benchmark both models using `torch.compile`. Interestingly, compiling the entire LLaMA model increases latency for the [Hugging Face](#) implementation, and compiling only the DyT or RMSNorm layers yields more efficient execution. Table 15 shows that, after compilation, the latency of the RMSNorm and DyT layers becomes nearly identical.

LLaMA 7B	inference		training	
	layer	model	layer	model
RMSNorm	0.3s	12.3s	3.9s	38.9s
DyT	0.3s	12.3s	3.9s	38.9s

Table 15 Inference and training latency (BF16 precision) for a LLaMA 7B with compiled RMSNorm or DyT. After compilation, the latency of RMSNorm and DyT layers are nearly identical.

An important distinction of DyT is that it is an element-wise operation and does not require a reduction operation within itself, compared to normalization layers. This could make it faster on hardware where reduction is a bottleneck. Additionally, even on conventional GPUs, DyT could offer opportunities for further optimization, e.g., fusing it with the preceding matrix multiplication layer from the last residual block.

D Replacing Batch Normalization with DyT

We investigate the potential of replacing BN with DyT in classic ConvNets such as ResNet-50 ([He et al., 2016](#)) and VGG19 ([Simonyan and Zisserman, 2014](#)). Both models are trained on the ImageNet-1K dataset ([Deng et al., 2009](#)) using the training recipes provided by `torchvision`. The DyT models are trained using the same hyperparameters as their BN counterparts.

model	BN	DyT
ResNet-50	76.2%	68.9%
VGG19	72.7%	71.0%

Table 16 ImageNet-1K classification accuracy with BN and DyT. Replacing BN with DyT in ResNet-50 and VGG19 results in a performance drop, indicating that DyT cannot fully substitute BN in these architectures.

The results are summarized in Table 16. Replacing BN with DyT led to a noticeable drop in classification accuracy for both models. These findings indicate that DyT is struggling to fully replace BN in these classic ConvNets. We hypothesize this could be related to BN layers being more frequent in these ConvNets, where they appear once with every weight layer, but LN only appears once per several weight layers in Transformers.